

Introduction to Business Intelligence & Power BI

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IS 107 - Business Intelligence

Part 1 | Step 1 : Open Power BI Desktop

Unlabeled - Power BI Desktop

Home

Open

Select a data source or start with a blank report

- Blank report
- OneLake catalog
- Excel workbook
- SQL Server
- Learn with sample data
- Get data from other sources

Recommended

Getting started

Intro—What is Power BI?

Recent Shared with me

Filter by keyword Filter

Name	Location	Opened
Project_Timeline.xlsx	C:\Users\Juser\Downloads\Project_Timeline.xlsx	5 days ago
effective_data.xlsx	C:\Users\Juser\Desktop\effective_data.xlsx	5 days ago

Sign in

Options and settings

About

Part 1 | Step 2: Load the Dataset

BI_LW1_Basic_Sales_Data (1).csv

File Origin: 1252: Western European (Windows) | Delimiter: Comma | Data Type Detection: Based on first 200 rows

Date	Product	Category	Region	Sales
01/01/2024	Product A	Electronics	East	175
02/01/2024	Product B	Furniture	South	178
03/01/2024	Product A	Electronics	North	842
04/01/2024	Product E	Furniture	North	582
05/01/2024	Product A	Electronics	South	245
06/01/2024	Product B	Furniture	North	667
07/01/2024	Product E	Furniture	West	353
08/01/2024	Product B	Furniture	East	609
09/01/2024	Product A	Electronics	West	813
10/01/2024	Product C	Office Supplies	South	434
11/01/2024	Product B	Furniture	North	494
12/01/2024	Product A	Electronics	North	539
13/01/2024	Product C	Office Supplies	East	502
14/01/2024	Product A	Electronics	West	897
15/01/2024	Product E	Furniture	West	277
16/01/2024	Product A	Electronics	East	715
17/01/2024	Product E	Furniture	South	520
18/01/2024	Product A	Electronics	South	196
19/01/2024	Product C	Office Supplies	South	232
20/01/2024	Product A	Electronics	East	539

The data in the preview has been truncated due to size limits.

Extract Table Using Examples | Load | Transform Data | Cancel

Part 1| Step 3 Verify Data in Data View

Untitled - Power BI Desktop

File Home View Table tools

Name: BLANK_Query_Sales

Workbook

Manage relationships

New Query Measure PivotTable PivotChart

Mark as data source

Date	Product	Category	Region	Sales
Sunday, 21 January 2024	Product C	Electronics	East	656
Friday, 2 February 2024	Product D	Electronics	West	606
Sunday, 11 February 2024	Product C	Electronics	West	760
Thursday, 15 February 2024	Product D	Electronics	West	311
Sunday, 3 March 2024	Product D	Electronics	West	679
Saturday, 16 March 2024	Product D	Electronics	West	512
Sunday, 17 March 2024	Product D	Electronics	North	629
Sunday, 24 March 2024	Product D	Electronics	West	627
Friday, 29 March 2024	Product D	Electronics	West	451
Saturday, 13 April 2024	Product C	Electronics	North	673
Friday, 19 April 2024	Product D	Electronics	East	736
Sunday, 3 May 2024	Product D	Electronics	North	286
Sunday, 12 May 2024	Product D	Electronics	South	663
Tuesday, 14 May 2024	Product D	Electronics	West	396
Saturday, 18 May 2024	Product D	Electronics	East	486
Saturday, 25 May 2024	Product D	Electronics	North	756
Sunday, 26 May 2024	Product D	Electronics	South	760
Sunday, 3 June 2024	Product D	Electronics	West	666
Thursday, 6 June 2024	Product D	Electronics	North	711
Saturday, 8 June 2024	Product D	Electronics	East	763
Sunday, 9 June 2024	Product D	Electronics	West	666
Sunday, 16 June 2024	Product D	Electronics	North	753
Monday, 17 June 2024	Product D	Electronics	South	556
Tuesday, 18 June 2024	Product D	Electronics	West	656
Monday, 24 June 2024	Product D	Electronics	East	663
Tuesday, 25 June 2024	Product D	Electronics	West	757
Thursday, 27 June 2024	Product D	Electronics	East	656
Thursday, 4 July 2024	Product D	Electronics	South	666
Saturday, 6 July 2024	Product D	Electronics	West	466
Sunday, 7 July 2024	Product D	Electronics	South	253
Monday, 8 July 2024	Product D	Electronics	North	546
Thursday, 11 July 2024	Product D	Electronics	South	576
Friday, 12 July 2024	Product D	Electronics	East	376
Saturday, 13 July 2024	Product D	Electronics	North	667
Sunday, 14 July 2024	Product D	Electronics	West	466
Wednesday, 17 July 2024	Product D	Electronics	North	756
Wednesday, 17 July 2024	Product D	Electronics	East	666

Table (18, 587) Data, Sales, Date (7:10:00 AM)

Are all columns visible?

Answer: Yes

Is “Date” formatted as Date?

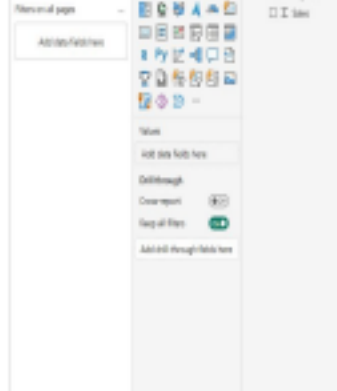
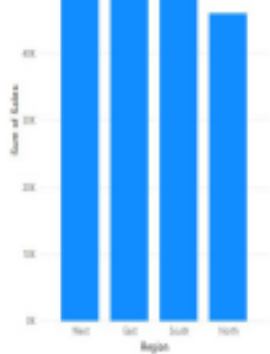
Answer: Yes

Is “Sales” formatted as Decimal Number?

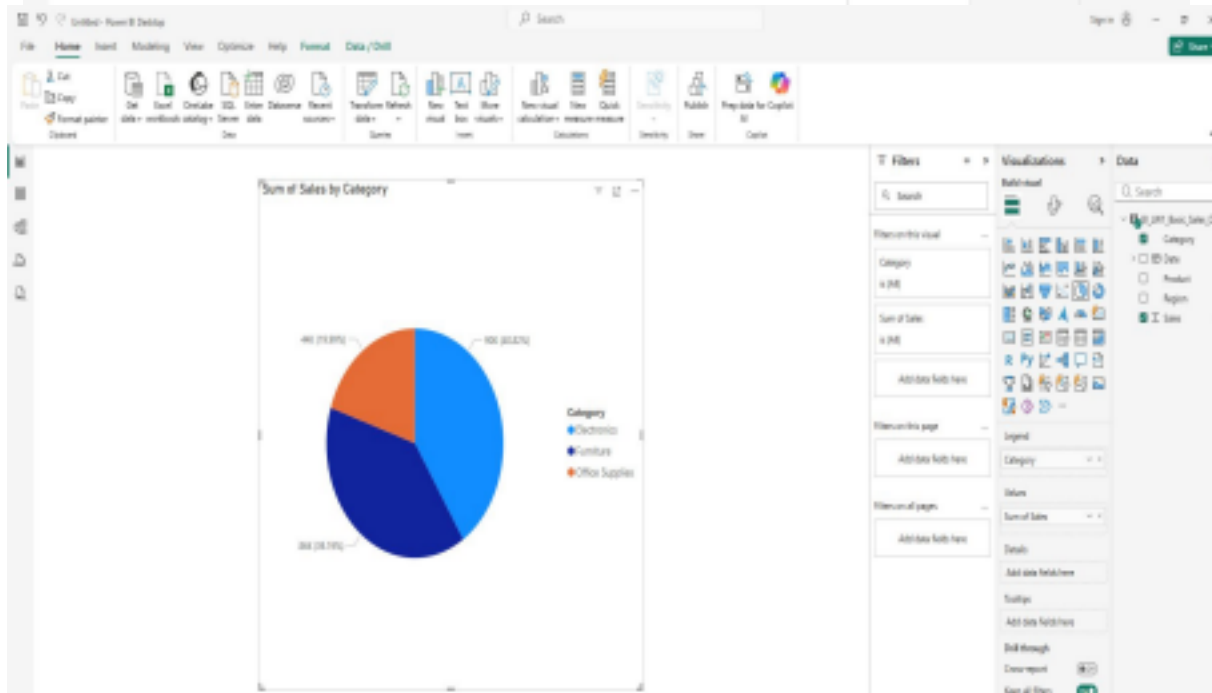
Answer: No

Creating Auto-Generated Visuals

Part 3 | Step 1 Quick Visualization



Which region has highest sales?



Answer: West Region
Creating Auto-Generated Visuals

Part 3 | Step 3 Sales by Category

Which category dominates?

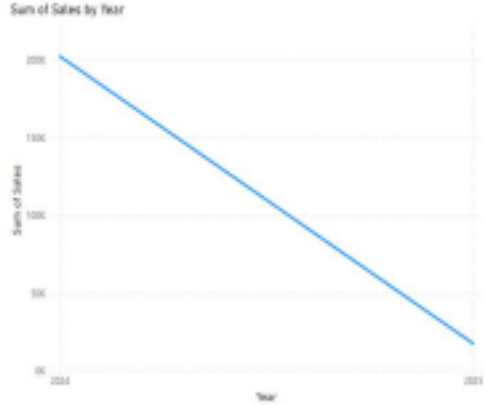
Answer: Electronics

Is the distribution balanced?

Answer: No

Creating Auto-Generated Visuals

Part 3 | Step 4 Sales by Category



Power BI Desktop interface showing the 'Fields' pane on the right. The 'Date' field is selected in the 'Filters on all pages' section. The 'Values' section shows 'Sum of Sales' as the aggregation. The 'Build Through' section shows 'Cross-report' and 'Keep all Rows' options. The 'Add still through fields here' section is empty.

Is there growth?

Answer: None

Any noticeable trend?

Answer: None

Basic Data Insight Interpretation

Which region contributes most revenue?

Answer: West

Which product category performs best?

Answer: Electronics

Are sales consistent across dates?

Answer: No

What business recommendation can you suggest?

Answer: Investing in North region.

Laboratory Questions Part A – Technical Questions

What are the five columns in the dataset?

Answer: Date, Product, Category, Region and Sales

What data type is assigned to the “Sales” column?

Answer: Whole Number

Which Power BI view allows you to see raw data?

Answer: Table View

What chart type is best for showing trends over time?

Answer: Line Chart

What aggregation is automatically applied to Sales?

Answer: SUM

Laboratory Questions

Part B – Technical Questions

Which region has the highest total sales?

Answer: West

Which category has the lowest performance?

Answer: Office Supplies

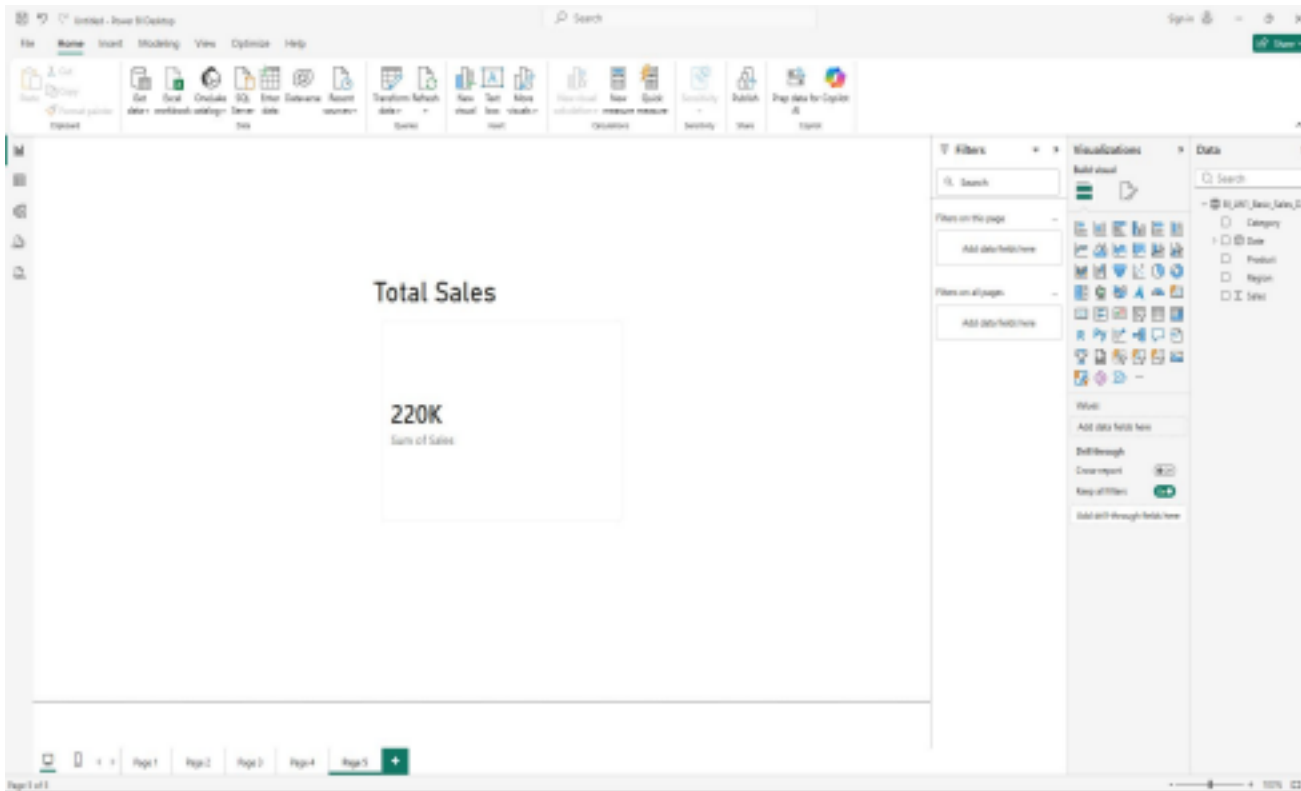
Are sales increasing, decreasing, or stable?

Answer: Decreasing

If you were a manager, which region would you prioritize?

Answer: North

Provide one actionable recommendation based on the data.



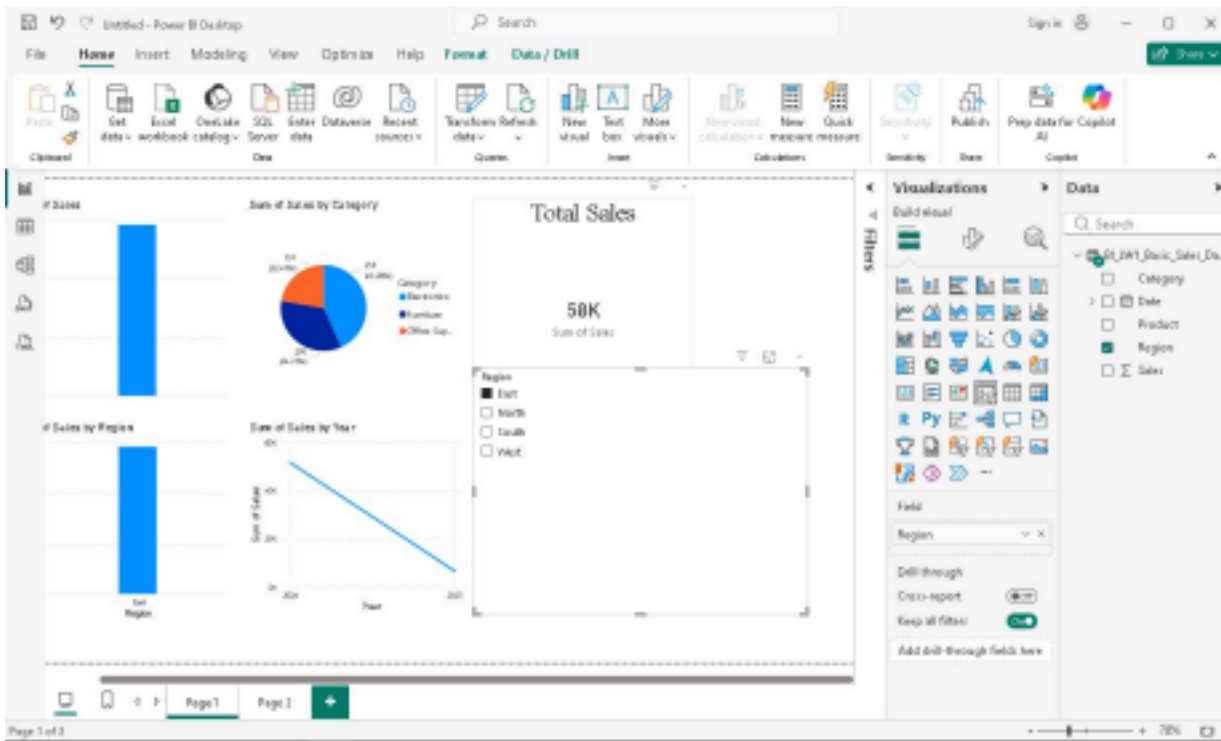
North region.

What is the total sales amount?

Answer:220k

ENHANCEMENT SECTION

Advanced Exploration



What happens to other visuals when you click a region?

Answer: The other visual will change.

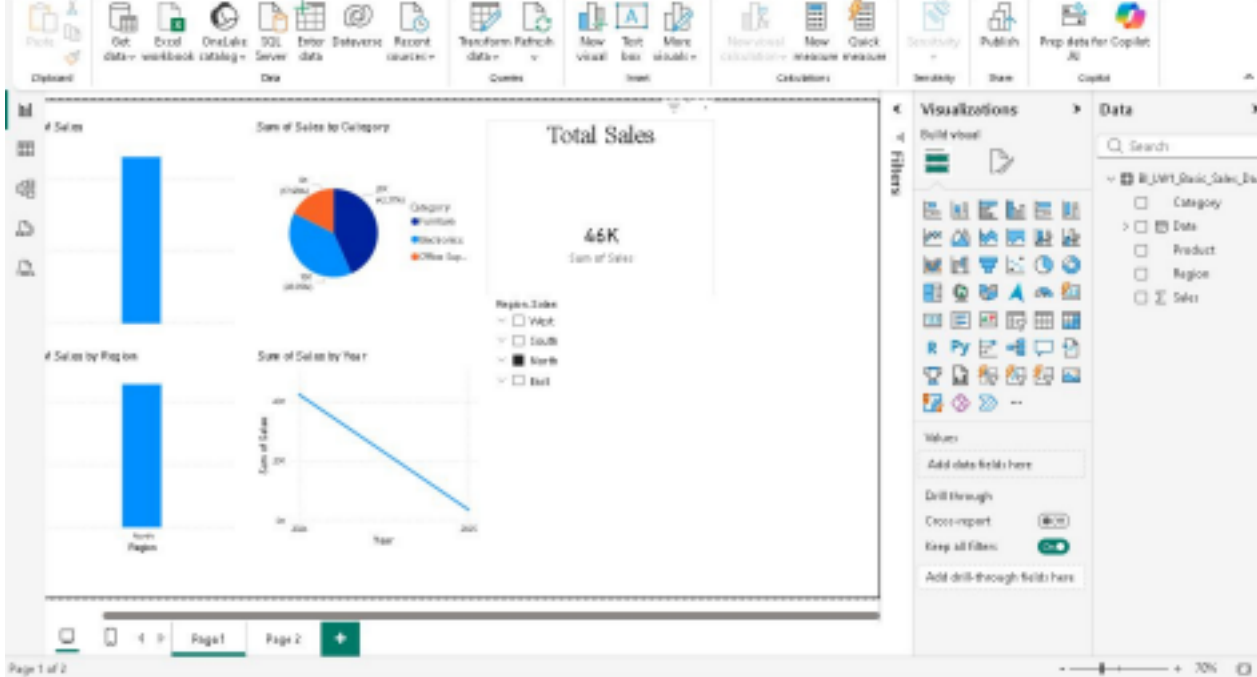
Why is filtering important in BI?

Answer: Filtering is important in BI because it transforms overwhelming data into actionable insights.

ENHANCEMENT SECTION

Advanced Exploration

Task 3 Sort Sales



Does sorting improve readability?

Answer: No

Why?

Answer: The sorting only rearrange the region in the card in descending order but the value still the same.

ENHANCEMENT SECTION

Advanced Exploration

Task 4 Identify Outliers

Which region is significantly higher or lower?

answer: The West region is higher and the North region is lower.

What might explain that difference?

answer: The sales has a big impact between this two region because it proves that the west region has a lot more buyers than in the north region.